

CDK12 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes CDK12 gene fusions from the msk_impact_50k_2026 study, covering 43 fusion events. 7/43 fusions (16.3%) were in-frame. 16/43 fusions (37.2%) retained the PF00069 domain. Domain retention was tested with Fisher's exact test ($p=0.0677006$, not statistically significant at $\alpha=0.05$). No literature benchmark is configured for CDK12.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 43 analyzed fusion events, identifying 29 distinct fusion partner genes. The most frequent partner was FBXL20, observed in 6 events.

Domain retention

PF00069 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 5/7 (71.4%) of in-frame fusions retained the domain, $p=0.0677006$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.316832.

Domain disruption

Domain-disruption analysis was skipped: CDK12 has no disruption_required_domains configured; domain-disruption analysis was skipped.

Cutpoint detection

Cutpoint detection scanned 41 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 644 aa (permutation-corrected $p=0.00990099$, statistically significant at $\alpha=0.05$). This is -84 aa from the nearest configured domain boundary at 728 aa.

Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: CDK12 has no group_field configured; confidence-stats analysis was skipped.

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention reported: No target exon was supplied in params or GeneConfig.

Joint partner

Joint partner reported: CDK12 has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ADCY9	2	0.046511627906976744	0.04991713954627366	None	1.0	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.25204681894094494	1
AKAP8L	1	0.023255813953488372	0.04991713954627366	None	0.9906291834002677	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.24007200940379297	2
HTR1E	2	0.046511627906976744	0.04991713954627366	None	0.9250334672021419	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.23598256191283248	3
MED13	1	0.023255813953488372	0.04991713954627366	None	0.9451137884872824	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.23031871049386757	4
ODAD4	1	0.023255813953488372	0.04991713954627366	None	0.9210174029451138	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.22515519930626002	5
TBKBP1	1	0.023255813953488372	0.04991713954627366	None	0.9210174029451138	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.22515519930626002	6
GRB7	1	0.023255813953488372	0.04991713954627366	None	0.85809906291834	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2116726978719513	7
ERBB2	3	0.06976744186046512	0.04991713954627366	None	0.7367246764837126	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.20559745559609263	8
RECQL4	1	0.023255813953488372	0.04991713954627366	None	0.7817938420348058	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.19532157911119402	9
STAT5B	4	0.09302325581395349	0.04991713954627366	None	0.5917001338688086	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.1844875453015368	10
COLEC12	1	0.023255813953488372	0.04991713954627366	None	0.6974564926372155	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.17724928995456754	11
CACNA1G	1	0.023255813953488372	0.04991713954627366	None	0.6613119143239625	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.16950402317315616	12
FBXL20	6	0.13953488372093023	0.04991713954627366	None	0.4152164212405176	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.16660316169846445	13
SYNRG	1	0.023255813953488372	0.04991713954627366	None	0.6265060240963856	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.16204561812438967	14

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ANKFN1	1	0.023255813953488372	0.04991713954627366	None	0.6050870147255689	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.1574558304020718	15
SKAP1	1	0.023255813953488372	0.04991713954627366	None	0.5917001338688086	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.15458721307562318	16
USP6	1	0.023255813953488372	0.04991713954627366	None	0.5917001338688086	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.15458721307562318	17
CSF3	1	0.023255813953488372	0.04991713954627366	None	0.5394912985274432	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.14339960550247344	18
FAM222B	1	0.023255813953488372	0.04991713954627366	None	0.4029451137884873	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.11413970877269718	19
IKZF3	2	0.046511627906976744	0.04991713954627366	None	0.321285140562249	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.10660792048999827	20
SPON1	1	0.023255813953488372	0.04991713954627366	None	0.32797858099062915	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.09807545174458471	21
SEC31A	1	0.023255813953488372	0.04991713954627366	None	0.2449799196787149	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.08029002432060309	22
VMP1	1	0.023255813953488372	0.04991713954627366	None	0.2449799196787149	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.08029002432060309	23
SNF8	1	0.023255813953488372	0.04991713954627366	None	0.18340026773761708	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.06709438461893927	24
MAP3K3	2	0.046511627906976744	0.04991713954627366	None	None	None	['domain_retention', 'recurrence']	0.04805958774302079	25
MTRES1	1	0.023255813953488372	0.04991713954627366	None	0.048192771084337394	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.03812134962180791	26
IGFBP4	1	0.023255813953488372	0.04991713954627366	None	0.01740294511378848	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.031523529770976	27
FGFR4	1	0.023255813953488372	0.04991713954627366	None	0.01338688085676043	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.030662944573041417	28
RPS6KB1	1	0.023255813953488372	0.04991713954627366	None	0.0	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.027794327246592757	29

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
80	1	15	0	25	None	0.39024390243902435	0.4086638740638108
240	2	14	0	25	None	0.14634146341463414	0.8346326063360919
246	3	13	0	25	None	0.052532833020637895	1.279569177684353
349	4	12	0	25	None	0.01797175866495507	1.7454094219943264
603	5	11	0	25	None	0.0058286784859313745	2.2344299000136965
637	6	10	0	25	None	0.0017809850929234755	2.749339715622759
644	9	7	0	25	None	3.265366098560993e-05	4.486068120461908
700	9	7	2	23	14.785714285714286	0.0011510086327457892	2.938921419081068
703	11	5	2	23	25.3	7.698377534159563e-05	4.113600794594795
750	11	5	3	22	16.133333333333333	0.0004274646628148892	3.369099781237133
807	11	5	4	21	11.55	0.0009913088581971262	3.003791013009668
870	11	5	5	20	8.8	0.003014843556286022	2.520735218981372
897	11	5	6	19	6.966666666666667	0.008600185164931898	2.0654921981468433
923	11	5	9	16	3.9111111111111111	0.05784571299099636	1.2377288215582707
949	11	5	15	10	1.4666666666666666	0.7420848641039097	0.12954642638599925
955	11	5	16	9	1.2375	1.0	-0.0
988	11	5	17	8	1.035294117647059	1.0	-0.0
1032	12	4	17	8	1.411764705882353	0.7341702326198091	0.13420322820990868
1090	12	4	18	7	1.1666666666666667	1.0	-0.0
1121	12	4	19	6	0.9473684210526315	1.0	-0.0
1146	13	3	19	6	1.368421052631579	1.0	-0.0
1159	13	3	20	5	1.0833333333333333	1.0	-0.0
1208	13	3	22	3	0.5909090909090909	0.6623956829348356	0.17888250666352729
1254	15	1	22	3	2.0454545454545454	1.0	-0.0
1355	16	0	22	3	None	0.26829268292682923	0.5713911715615105
1378	16	0	23	2	None	0.5121951219512195	0.2905645619858162
1381	16	0	24	1	None	1.0	-0.0

domain_retention tables

frame_domain_contingency_table

5	11
2	23

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	41	16	15	10	0.22484641638225256

frequency tables

Partner_gene_counts

Partner_gene	Event_count
ADCY9	2
AKAP8L	1
ANKFN1	1
CACNA1G	1
COLEC12	1
CSF3	1
ERBB2	3
FAM222B	1
FBXL20	6
FGFR4	1
GRB7	1
HTR1E	2
IGFBP4	1
IKZF3	2
MAP3K3	2
MED13	1
MTRES1	1
ODAD4	1
RECQL4	1
RPS6KB1	1
SEC31A	1
SKAP1	1
SNF8	1
SPON1	1
STAT5B	4
SYNRG	1
TBKBP1	1
USP6	1

Partner_gene	Event_count
VMP1	1

Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003 9125-T01-I M6-1	P-0039 125-T0 1-IM6		AKAP 8L-CD K12	AKA P8L	unk now n	thr ee _pr ime	376279 94	2	637	False	retai ned	1.0	False	Protein kinase domai n			AKAP8L (NM_014371) - CDK12 (NM_016507) rearrangement: t(17;19)(q1 2;p13.12)(chr17:g.376279 94::chr19:g.15529663) Protein Fusion: mid-exon {AKAP8L:CDK12}	NA
EVT-P-005 6622-T01-I M6-2	P-0056 622-T0 1-IM6		ANKF N1-CD K12	ANK FN1	in-fr ame	thr ee _pr ime	376193 99	1	349	True	retai ned	1.0	False	Protein kinase domai n			ANKFN1 (NM_153228) - CDK12 (NM_016507) rearrangement: c.62+8631 :ANKFN1_c.1046+29:CDK 12dup Protein Fusion: in frame {ANKFN1:CDK12}	NA
EVT-P-000 3762-T02-I M5-4	P-0003 762-T0 2-IM5		CDK1 2-ADC Y9	ADC Y9	unk now n	thr ee _pr ime	376466 65	3	644	True	retai ned	1.0	False	Protein kinase domai n			ADCY9 (NM_001116) - CDK12 (NM_016507) rearrangement: t(16;17)(p1 3.3;q12)(chr16:g.4033430: :chr17:g.37646665) Protein fusion: mid-exon (ADCY9-CDK12)	NA
EVT-P-000 3762-T01-I M5-5	P-0003 762-T0 1-IM5		CDK1 2-ADC Y9	ADC Y9	unk now n	thr ee _pr ime	376466 65	3	644	True	retai ned	1.0	False	Protein kinase domai n			t(16;17)(p13.3;q12)(chr16: g.4033430::chr17:g.37646 665) Protein fusion: mid-exon (ADCY9-CDK12)	NA
EVT-P-002 4871-T01-I M6-7	P-0024 871-T0 1-IM6		CDK1 2-CAC NA1G	CAC NA1 G	unk now n	five _pr ime	376678 06	8	897	False	disru pted	0.5802047 781569966	True			Protein kinase domain	CDK12 (NM_016507) - CACNA1G (NM_018896) rearrangement: c.2691:CD K12_c.6399+226:CACNA1 Gdel Protein Fusion: mid-exon {CDK12:CACNA1G}	NA
EVT-P-003 8990-T01-I M6-53	P-0038 990-T0 1-IM6		CDK1 2-FBX L20	FBX L20	in-fr ame	five _pr ime	376763 67	11	1032	True	retai ned	1.0	False	Protein kinase domai n			CDK12 (NM_016507) - FBXL20 (NM_032875) rearrangement: c.3095+27 :CDK12_c.104+18457:FB XL20inv Protein Fusion: in frame {CDK12:FBXL20}	NA
EVT-P-000 2174-T01-I M3-55	P-0002 174-T0 1-IM3		CDK1 2-GRB 7	GR B7	unk now n	thr ee _pr ime	376507 68	5	750	True	disru pted	0.9249146 757679181	True			Protein kinase domain	NA Transcript fusion (GRB7-CDK12)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intron_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008960-T02-I M6-56	P-0008960-T02-IM6		CDK12-HTR1E	HTR1E	unknown	five_prime	37646977	3	700	False	lost	0.0	False		Protein kinase domain		CDK12 (NM_016507) - HTR1E (NM_000865) rearrangement: t(6;17)(q14.3;q12)(chr6:g.87697113::chr17:g.37646977) Protein Fusion: mid-exon {CDK12:HTR1E}	NA
EVT-P-0008960-T01-I M5-57	P-0008960-T01-IM5		CDK12-HTR1E	HTR1E	unknown	five_prime	37646977	3	700	False	lost	0.0	False		Protein kinase domain		CDK12 (NM_016507) - HTR1E (NM_000865) rearrangement: t(6;17)(q14.3;q12)(chr6:g.87697112::chr17:g.37646977) Protein fusion: mid-exon (CDK12-HTR1E)	NA
EVT-P-0060111-T01-I M7-58	P-0060111-T01-IM7		CDK12-IKZF3	IKZF3	in-frame	five_prime	37682742	13	1254	True	retained	1.0	False	Protein kinase domain			CDK12 (NM_016507) - IKZF3 (NM_012481) rearrangement: c.3760+173:CDK12_c.8-12056:IKZF3inv Protein Fusion: in frame {CDK12:IKZF3}	NA
EVT-P-0040752-T01-I M6-62	P-0040752-T01-IM6		CDK12-MTR ES1	MTR ES1	unknown	five_prime	37687160	14	1355	False	retained	1.0	False	Protein kinase domain			CDK12 (NM_016507) - C6orf203 (NM_016487) rearrangement: t(6;17)(q21.3;q12)(chr6:g.107368450::chr17:g.37687160) Protein Fusion: mid-exon {CDK12:C6orf203}	NA
EVT-P-0009835-T01-I M5-67	P-0009835-T01-IM5		CDK12-RECQL4	RECQL4	out-of-frame	five_prime	37656296	6	807	True	disrupted	0.27303754266211605	True			Protein kinase domain	CDK12 (NM_016507) - RECQL4 (NM_004260) rearrangement: t(8;17)(q24.3;q12)(chr8:g.145739493::chr17:g.37656296) Protein fusion: out of frame (CDK12-RECQL4)	NA
EVT-P-0023550-T01-I M6-68	P-0023550-T01-IM6		CDK12-RPS6KB1	RPS6KB1	unknown	three_prime	37687269	14	1391	False	lost	0.0	False		Protein kinase domain		CDK12 (NM_016507) rearrangement: c.4173_chr17:g.58028299dup Protein Fusion: mid-exon {RPS6KB1:CDK12}	NA
EVT-P-0041371-T01-I M6-69	P-0041371-T01-IM6		CDK12-SKAP1	SKAP1	out-of-frame	five_prime	37673680	10	949	True	disrupted	0.757679180887372	True			Protein kinase domain	CDK12 (NM_016507) - SKAP1 (NM_003726) rearrangement: c.2847-13:CDK12_c.280+66555:SKAP1inv Protein Fusion: out of frame {CDK12:SKAP1}	NA
EVT-P-0043326-T01-I M6-71	P-0043326-T01-IM6		CDK12-SNF8	SNF8	out-of-frame	five_prime	37683411	13	1254	True	retained	1.0	False	Protein kinase domain			CDK12 (NM_016507) - SNF8 (NM_007241) rearrangement: c.3760+842:CDK12_c.422+1405:SNF8inv Protein Fusion: out of frame {CDK12:SNF8}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0057341-T01-IM6-73	P-0057341-T01-IM6		CDK12-SPO N1	SPO N1	unknown	five_prime	37682247	13	1146	False	retained	1.0	False	Protein kinase domain			CDK12 (NM_016507) - SPON1 (NM_006108) rearrangement: t(11;17)(p15.2;q12)(chr11:g.14149386::chr17:g.37682247) Protein Fusion: mid-exon {CDK12:SPON1}	NA
EVT-P-0016507-T04-IM6-74	P-0016507-T04-IM6		CDK12-STAT5B	STAT5B	unknown	three_prime	37672113	9	949	True	disrupted	0.24573378839590443	True			Protein kinase domain	CDK12 (NM_016507) - STAT5B (NM_012448) rearrangement: c.2846+52:CDK12_c.2238-225:STAT5Bdel Antisense Fusion	NA
EVT-P-0016507-T03-IM6-75	P-0016507-T03-IM6		CDK12-STAT5B	STAT5B	unknown	three_prime	37672113	9	949	True	disrupted	0.24573378839590443	True			Protein kinase domain	CDK12 (NM_016507) - STAT5B (NM_012448) rearrangement: c.2846+52:CDK12_c.2238-225:STAT5Bdel. Antisense Fusion	NA
EVT-P-0016507-T02-IM6-76	P-0016507-T02-IM6		CDK12-STAT5B	STAT5B	unknown	three_prime	37672113	9	949	True	disrupted	0.24573378839590443	True			Protein kinase domain	CDK12 (NM_016507) - STAT5B (NM_012448) rearrangement: c.2846+52:CDK12_c.2238-225:STAT5Bdel Antisense Fusion	NA
EVT-P-0016507-T01-IM6-77	P-0016507-T01-IM6		CDK12-STAT5B	STAT5B	unknown	three_prime	37672113	9	949	True	disrupted	0.24573378839590443	True			Protein kinase domain	CDK12 (NM_016507) - STAT5B (NM_012448) rearrangement: c.2846+52:CDK12_c.2238-225:STAT5Bdel Antisense Fusion	NA
EVT-P-0023516-T01-IM6-78	P-0023516-T01-IM6		COLEC12-CDK12	COLEC12	out-of-frame	three_prime	37665943	7	870	True	disrupted	0.515358361774744	True			Protein kinase domain	COLEC12 (NM_130386) - CDK12(NM_016507) rearrangement: t(17;18)(q12;p11.32)(chr17:g.37665943::chr18:g.386902) Protein Fusion: out of frame {COLEC12:CDK12}	NA
EVT-P-0038406-T01-IM6-80	P-0038406-T01-IM6		CSF3-CDK12	CSF3	out-of-frame	three_prime	37676083	11	988	True	disrupted	0.11262798634812286	True			Protein kinase domain	CSF3 (NM_000759) - CDK12 (NM_016507) Rearrangement : c.313-58:CSF3_c.2964-126:CDK12 dup Protein Fusion: out of frame {CSF3:CDK12}	NA
EVT-P-0046790-T01-IM6-81	P-0046790-T01-IM6		ERBB2-CDK12	ERBB2	out-of-frame	three_prime	37640037	3	644	True	retained	1.0	False	Protein kinase domain			ERBB2 (NM_004448) - CDK12 (NM_016507) rearrangement: c.225+207:ERBB2_c.1932-6773:CDK12dup Protein Fusion: out of frame {ERBB2:CDK12}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intron_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004 3706-T01-I M6-82	P-0043 706-T0 1-IM6		ERBB2-CDK12	ERBB2	unknown	threonine_prime	37668120	8	923	True	disrupted	0.33447098976109213	True			Protein kinase domain	ERBB2 (NM_004448) - CDK12 (NM_016507) rearrangement: c.2293:ERBB2_c.2768+237:CDK12dup Protein Fusion: mid-exon {ERBB2:CDK12}	NA
EVT-P-003 7167-T01-I M6-83	P-0037 167-T0 1-IM6		ERBB2-CDK12	ERBB2	unknown	threonine_prime	37673710	10	955	False	disrupted	0.22525597269624573	True			Protein kinase domain	ERBB2 (NM_004448) - CDK12 (NM_016507) rearrangement: c.2553:ERBB2_c.2864:CDK12dup Protein Fusion: mid-exon {ERBB2:CDK12}	NA
EVT-P-003 6357-T01-I M6-84	P-0036 357-T0 1-IM6		FAM22B-CDK12	FAM22B	unknown	threonine_prime	37681100	12	1090	False	lost	0.0	False		Protein kinase domain		CDK12 (NM_016507) rearrangement: c.3269:CDK12_chr17:g.27096697inv Protein Fusion: mid-exon {FAM22B:CDK12}	NA
EVT-P-001 4403-T01-I M6-86	P-0014 403-T0 1-IM6		FBXL20-CDK12	FBXL20	unknown	threonine_prime	37682171	13	1121	False	lost	0.0	False		Protein kinase domain		FBXL20 (NM_032875) - CDK12 (NM_016507) rearrangement: c.43-9338:FBXL20_c.3362:CDK12inv Protein fusion: mid-exon (FBXL20-CDK12)	NA
EVT-P-004 6675-T01-I M6-87	P-0046 675-T0 1-IM6		FBXL20-CDK12	FBXL20	in-frame	threonine_prime	37671717	9	923	True	disrupted	0.33447098976109213	True			Protein kinase domain	FBXL20 (NM_032875) - CDK12 (NM_016507) rearrangement: c.398+4667:FBXL20_c.2769-267:CDK12inv Protein Fusion: in frame {FBXL20:CDK12}	NA
EVT-P-004 3910-T01-I M6-88	P-0043 910-T0 1-IM6		FBXL20-CDK12	FBXL20	unknown	threonine_prime	37682285	13	1159	False	lost	0.0	False		Protein kinase domain		FBXL20 (NM_032875) - CDK12 (NM_016507) rearrangement: c.42+20635:FBXL20_c.3476:CDK12inv Protein Fusion: mid-exon {FBXL20:CDK12}	NA
EVT-P-003 6553-T01-I M6-89	P-0036 553-T0 1-IM6		FBXL20-CDK12	FBXL20	unknown	threonine_prime	37619062	1	246	False	retained	1.0	False	Protein kinase domain			FBXL20 (NM_032875) - CDK12 (NM_016507) rearrangement: c.697-2344:FBXL20_c.738:CDK12inv Protein Fusion: mid-exon {FBXL20:CDK12}	NA
EVT-P-004 8892-T01-I M6-90	P-0048 892-T0 1-IM6		FBXL20-CDK12	FBXL20	unknown	threonine_prime	37682432	13	1208	False	lost	0.0	False		Protein kinase domain		FBXL20 (NM_032875) - CDK12 (NM_016507) rearrangement: c.42+25196:FBXL20_c.3623:CDK12inv Protein Fusion: mid-exon {FBXL20:CDK12}	NA

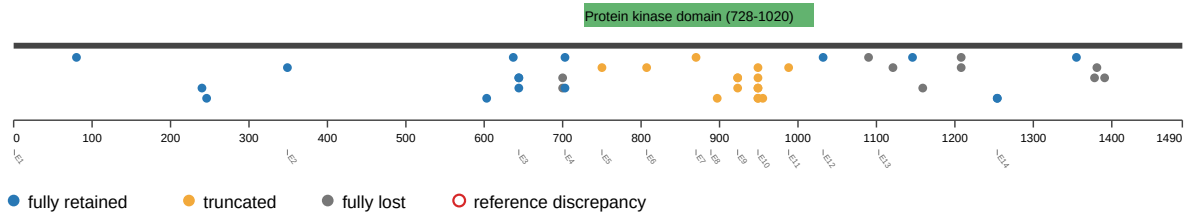
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intron_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002 6288-T03-I M6-91	P-0026 288-T0 3-IM6		FGFR 4-CDK 12	FGFR4	unknown	three_prime	37687238	14	1381	False	lost	0.0	False		Protein kinase domain		FGFR4 (NM_213647) - CDK12 (NM_016507) rearrangement: t(5;17)(q35.2;q12)(chr5:g.176518227:chr17:g.37687238) Protein Fusion: mid-exon {FGFR4:CDK12}	NA
EVT-P-006 0015-T01-I M7-92	P-0060 015-T0 1-IM7		IGFBP4-CDK12	IGFBP4	unknown	three_prime	37687228	14	1378	False	lost	0.0	False		Protein kinase domain		IGFBP4 (NM_001552) - CDK12 (NM_016507) rearrangement: c.349+3777:IGFBP4_c.4132:CDK12dup Protein Fusion: mid-exon {IGFBP4:CDK12}	NA
EVT-P-006 8695-T01-I M7-93	P-0068 695-T0 1-IM7		IKZF3-CDK12	IKZF3	unknown	three_prime	37619043	1	240	False	retained	1.0	False	Protein kinase domain			IKZF3 (NM_012481) - CDK12 (NM_016507) rearrangement: c.8-3620:IKZF3_c.719:CDK12inv Protein Fusion: mid-exon {IKZF3:CDK12}	NA
EVT-P-004 8104-T01-I M6-94	P-0048 104-T0 1-IM6		MED13-CDK12	MED13	unknown	three_prime	37627894	2	603	False	retained	1.0	False	Protein kinase domain			MED13 (NM_005121) - CDK12 (NM_016507) rearrangement: c.2264-2554:MED13_c.1809:CDK12inv Protein Fusion: mid-exon {MED13:CDK12}	NA
EVT-P-006 5545-T01-I M7-95	P-0065 545-T0 1-IM7		ODAD4-CDK12	ODAD4	in-frame	three_prime	37648612	4	703	True	retained	1.0	False	Protein kinase domain			TTC25 (NM_031421) - CDK12 (NM_016507) rearrangement: c.1059-1499:TTC25_c.2109-392:CDK12dup Protein Fusion: in frame {TTC25:CDK12}	NA
EVT-P-004 8892-T01-I M6-97	P-0048 892-T0 1-IM6		SEC31A-CDK12	SEC31A	unknown	three_prime	37682432	13	1208	False	lost	0.0	False		Protein kinase domain		SEC31A (NM_014933) - CDK12 (NM_016507) rearrangement: t(4;17)(q21.22;q12)(chr4:g.83761890:chr17:g.37682432) Protein Fusion: mid-exon {SEC31A:CDK12}	NA
EVT-P-000 5530-T02-I M5-98	P-0005 530-T0 2-IM5		SYNRG-CDK12	SYNRG	out-of-frame	three_prime	37671800	9	923	True	disrupted	0.33447098976109213	True			Protein kinase domain	SYNRG (NM_007247) - CDK12 (NM_016507) rearrangement : c.118+961:SYNRG_c.2769-184:CDK12inv Protein fusion: out of frame (SYNRG-CDK12)	NA
EVT-P-004 1371-T01-I M6-99	P-0041 371-T0 1-IM6		TBKBP1-CDK12	TBKBP1	in-frame	three_prime	37648975	4	703	True	retained	1.0	False	Protein kinase domain			TBKBP1 (NM_014726) - CDK12 (NM_016507) rearrangement: c.872+1376:TBKBP1_c.2109-29:CDK12dup Protein Fusion: in frame {TBKBP1:CDK12}	NA

event_id	sampl e_id	pa tie nt _i d	fusion _nam e	part ner _ge ne	fra me _sta tus	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost _do mai ns	disrupt ed_do mains	source_annotation_text	source_site2 _effect_on_f rame
EVT-P-001 9776-T01-I M6-100	P-0019 776-T0 1-IM6		USP6- CDK1 2	USP 6	in-fr ame	thr ee _pr ime	376735 92	10	949	True	disru pted	0.2457337 883959044 3	True			Protein kinase domain	USP6 (NM_004505) - CDK12 (NM_016507) rearrangement: c.2916-47 0:USP6_c.2847-101:CDK1 2del Protein Fusion: in frame {USP6:CDK12}	NA
EVT-P-005 9578-T01-I M7-101	P-0059 578-T0 1-IM7		VMP1- CDK1 2	VM P1	unk now n	thr ee _pr ime	376185 64	1	80	False	retai ned	1.0	False	Protein kinase domai n			VMP1 (NM_030938) - CDK12 (NM_016507) rearrangement: c.582+174 7:VMP1_c.240:CDK12dup Protein Fusion: mid-exon {VMP1:CDK12}	NA

Figures

domain retention outliers

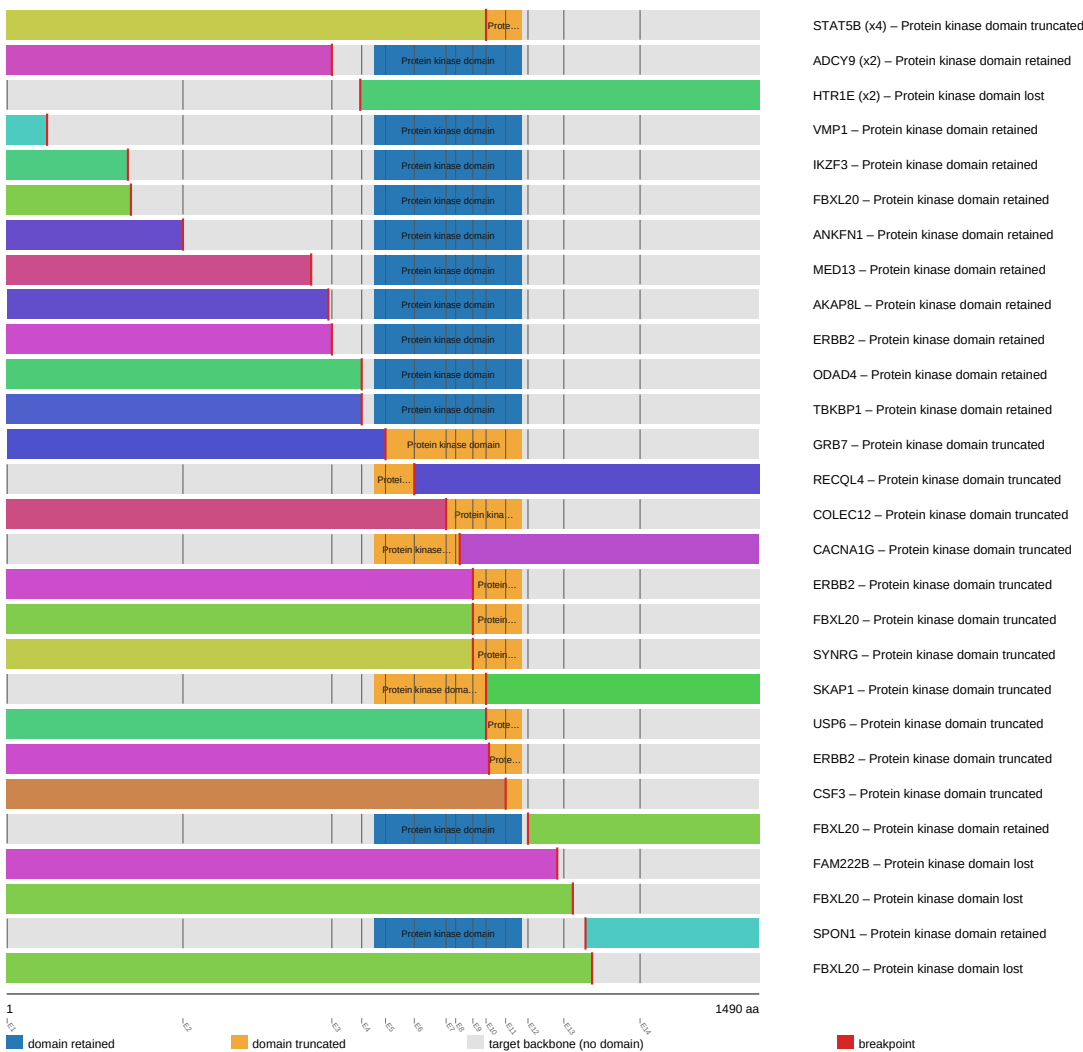
CDK12 domain-retention track



fusion schematic

CDK12 fusion-transcript schematic

partner block → breakpoint → retained CDK12 portion (domain-colored, exon ticks)



Showing the top 28 of 36 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

reference comparison

Reference vs msk_impact_50k_2026

